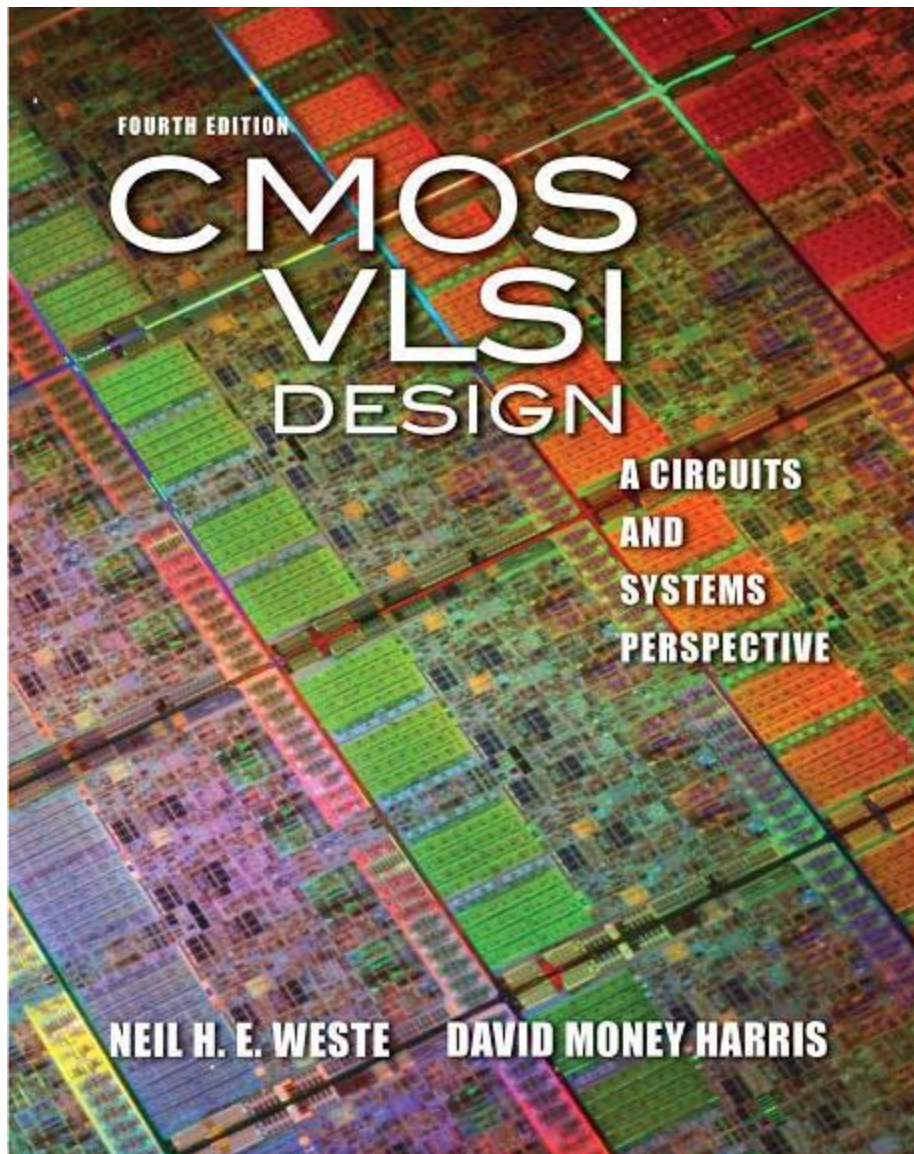
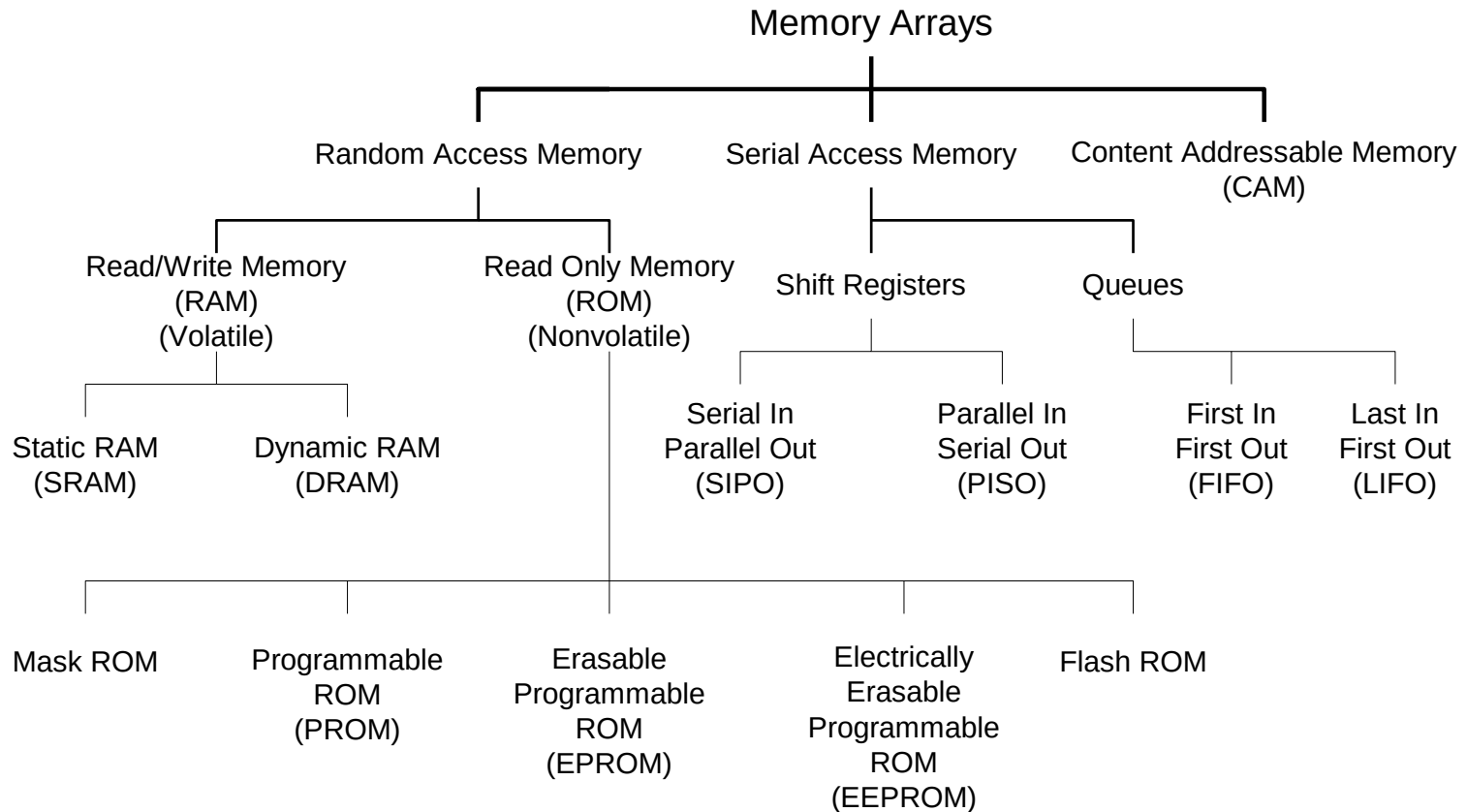




Array Subsystems

Memory Arrays



Lecture # 6

- ❑ Dynamic Random Access Memory (DRAM)
- ❑ DRAM Technology
- ❑ DRAM implementing the capacitor
- ❑ DRAM chip organization

Dynamic RAM

- ❑ DRAM used for main memory while SRAM used for cache
- ❑ DRAMs store their contents as charge on a capacitor rather than in a feedback loop
- ❑ DRAM basic cell is substantially smaller than SRAM
- ❑ DRAM cell must be periodically read and refreshed so that its contents do not leak away
- ❑ Commercial DRAMs are built in specialized processes optimized for dense capacitor structures.
- ❑ DRAMs offer a factor of 10–20 greater density (bits/cm²) than high-performance SRAM built in a standard logic process
- ❑ DRAM also have much higher latency.

DRAM Technology

❑ DRAM

- Must be re-written after being read
- Must also be periodically refreshed
 - Every ~ 8 ms
 - Each row can be refreshed simultaneously
- One transistor/bit, so higher density as compared to SRAM
- Slower compared to SRAM, used in main memory

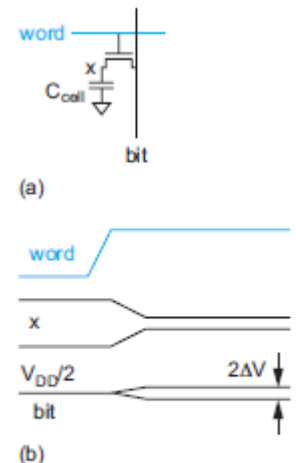
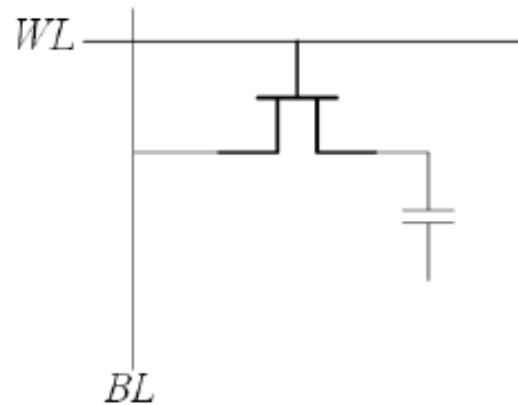


FIGURE 12.41
1T1R DRAM cell read operation

DRAM implementing the capacitor

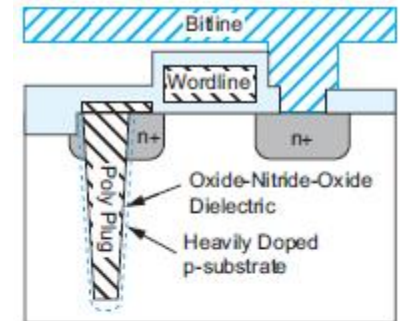
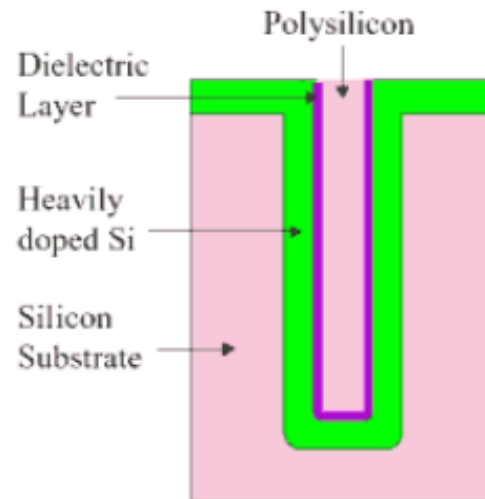
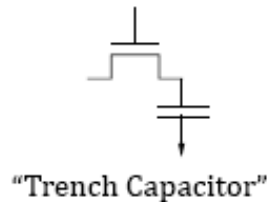
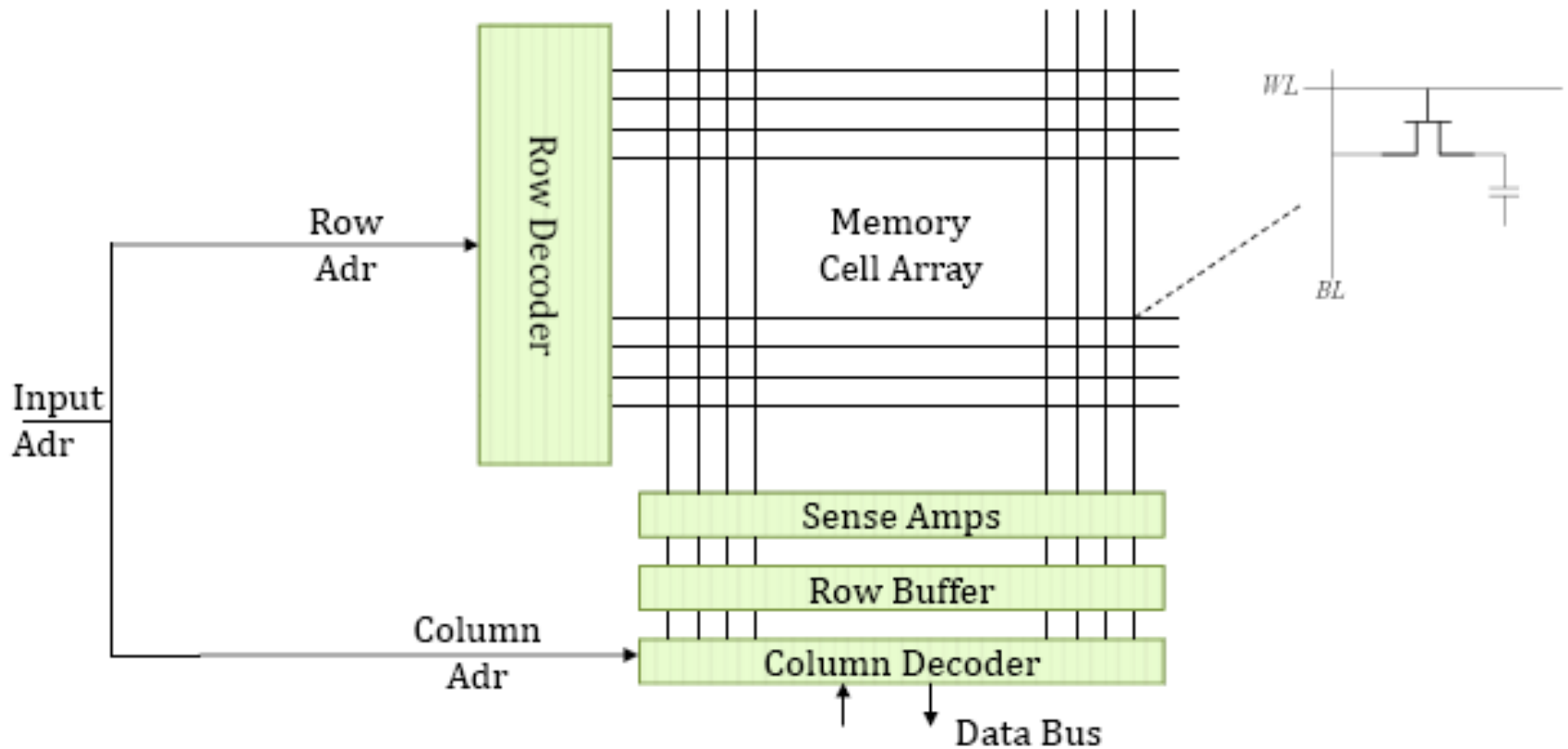


FIGURE 12.42 Trench capacitor

DRAM chip organization



DRAM chip organization

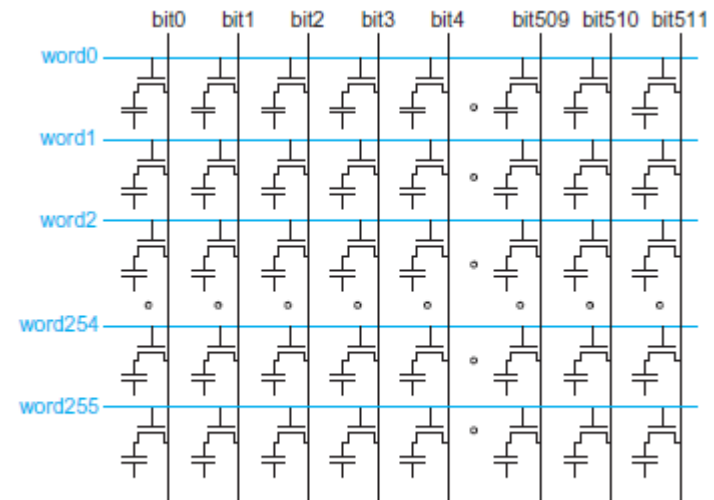
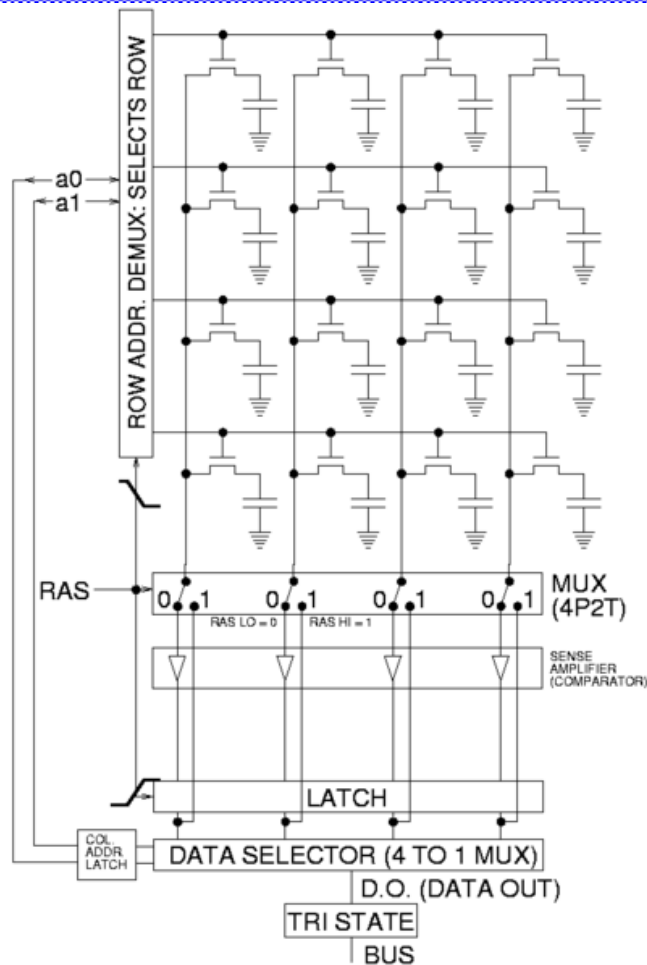


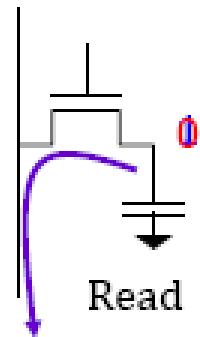
FIGURE 12.43 DRAM subarray

DRAM chip organization

- ❑ The addresses are multiplexed
 - Cuts down total address pins into half
- ❑ First send row address during Row Access Strobe (RAS)
 - It stores complete row to row buffer
- ❑ Then send column address during column address strobe (CAS)
- ❑ Row buffer
 - Read lots of bits all at once, and then send them out based on different column addresses
 - Similar to reading a full cache line, but only accessing one word at a time

DRAM refresh

- ❑ Reads are destructive. Contents are erased after reading
 - Charge stored in the capacitor is depleted
- ❑ After read, the contents of DRAM cell are gone
- ❑ The values are stored in the row buffer
- ❑ Write them back to the cells for future reference
- ❑ DRAM cell loses its value even if not accessed.
 - That's why it is called dynamic
 - All rows need to be refreshed at regular intervals through memory controller
 - Refresh process could yield memory unavailable occasionally



Memory technology

❑ Amdahl:

- Memory capacity should grow linearly with processor speed
- Unfortunately, memory capacity and speed has not kept pace with processors

❑ Some optimizations:

- Multiple accesses to same row
- Synchronous DRAM
 - Originally asynchronous
 - Added clock to DRAM interface to overcome synchronization overhead
 - Burst mode with critical word first
- Wider interfaces
- Double data rate (DDR)
- Multiple banks on each DRAM device

Memory optimizations

Production year	Chip size	DRAM type	Row access strobe (RAS)		Column access strobe (CAS)/ data transfer time (ns)	Cycle time (ns)
			Slowest DRAM (ns)	Fastest DRAM (ns)		
1980	64K bit	DRAM	180	150	75	250
1983	256K bit	DRAM	150	120	50	220
1986	1M bit	DRAM	120	100	25	190
1989	4M bit	DRAM	100	80	20	165
1992	16M bit	DRAM	80	60	15	120
1996	64M bit	SDRAM	70	50	12	110
1998	128M bit	SDRAM	70	50	10	100
2000	256M bit	DDR1	65	45	7	90
2002	512M bit	DDR1	60	40	5	80
2004	1G bit	DDR2	55	35	5	70
2006	2G bit	DDR2	50	30	2.5	60
2010	4G bit	DDR3	36	28	1	37
2012	8G bit	DDR3	30	24	0.5	31

Memory optimizations

- ❑ DDR:
 - DDR2
 - Lower power (2.5 V -> 1.8 V)
 - Higher clock rates (266 MHz, 333 MHz, 400 MHz)
 - DDR3
 - 1.5 V
 - 800 MHz
 - DDR4
 - 1-1.2 V
 - 1600 MHz
- ❑ Dual In-line Memory Module (DIMM)
 - DIMMs are labeled by peak bandwidth
 - PC2100 = 2x8x133 MHz
- ❑ GDDR5 is graphics memory based on DDR3

Memory optimizations

Standard	Clock rate (MHz)	M transfers per second	DRAM name	MB/sec /DIMM	DIMM name
DDR	133	266	DDR266	2128	PC2100
DDR	150	300	DDR300	2400	PC2400
DDR	200	400	DDR400	3200	PC3200
DDR2	266	533	DDR2-533	4264	PC4300
DDR2	333	667	DDR2-667	5336	PC5300
DDR2	400	800	DDR2-800	6400	PC6400
DDR3	533	1066	DDR3-1066	8528	PC8500
DDR3	666	1333	DDR3-1333	10,664	PC10700
DDR3	800	1600	DDR3-1600	12,800	PC12800
DDR4	1066–1600	2133–3200	DDR4-3200	17,056–25,600	PC25600

Memory optimizations

- ❑ Graphics memory:
 - Achieve 2-5 X bandwidth per DRAM vs. DDR3
 - Wider interfaces (32 vs. 16 bit)
 - Higher clock rate
 - Possible because they are connected directly to GPUs
- ❑ Reducing power in SDRAMs:
 - Both dynamic power and static power depend on operating voltage
 - Low power mode (ignores clock, continues to refresh)

Future trends in DDR

- ❑ Low power (LP) DDR2 or m-DDR2 (mobile)
- ❑ Used in PMDs or tablets
 - Apple iPad, Samsung Galaxy tab
- ❑ Consume low power
 - Reduced VDDs: 1.8 v, 1.2 v
 - Temperature-compensated refresh
 - Deep power down mode where RAM loses its data
- ❑ Smaller in size, less on board area

DRAM technology roadmap

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Process	3x nm		2x_H		2x_L		1x_H		1x_M	
DDR3	1600		1866							
DDR3L	1333		1600							
DDR4			1866	2133	2400	2667				
DDR4L						2400	2667	2667	2932	3200
Device		2Gb								
Device				4Gb						
Device							8Gb			
Device										16Gb
DIMM	8GB	16GB		32GB		64GB	64GB			128GB
3DS/TSV				DDR4_2H	DDR4_4H	DDR4_8H				

Anticipated roadmap by JEDEC (Joint Electron Device Engineering Council)

Future exotic memory technologies

- ❑ Memristors
 - A memristor changes its resistance according to the current passed through it
 - It then retains the resistance even if the voltage is removed
 - Can work as non-volatile memory
 - Program/erase depends upon the direction of current flow
- ❑ PCRAM (Phase Change RAM)
 - Uses unique properties of chalcogenide glass
 - Two states of glass (amorphous/crystalline)
 - Two states represent logic 0 or 1
 - New materials are being searched for more intermediate states
 - May be more bits can be stored per unit element

Memory dependability

- ❑ Memory is susceptible to cosmic rays
- ❑ *Soft errors*: dynamic errors
 - Can be detected by parity bits. Normally overhead of 1 parity bit per 8 bit data. Can detect only a single bit error. Multiple errors can not be detected
 - Detected and fixed by error correcting codes (ECC). 8 bit overhead for 64 bit data for two error detections and one error correction.
- ❑ *Hard errors*: permanent errors
 - Permanent errors in the function of memory cell
- ❑ Chipkill: a RAID-like error recovery technique
 - Chipkill distributes the data and ECC cross an array of memory chips.
 - Complete failure of memory chip can be handled with the help of data from other chips and ECC